CLASS 8 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If a pesticide kills all the frogs in a field, the number of insects will most likely: 1

(a) become constant **(b)** stay zero **(c)** increase **(d)** decrease

2. A student says the Earth's inner core is liquid because it is hotter than the outer core. This reasoning is wrong because: 1

(a) the inner core is gas **(b)** the outer core is cooler

(c) pressure raises the melting point **(d)** iron cannot be solid

3. A convex mirror is fitted in shops to see a wide area because it forms: 1

(a) no image

(b) an inverted image of the same size

(c) an enlarged real image

(d) an erect, diminished image that covers a wide field

4. After 33 years the lunar and solar years differ by about: 1

(a) ten years **(b)** one month **(c)** one year **(d)** one day

5. Lichen is an association of: 1

(a) algae and animals **(b)** algae and fungi

(c) fungi and animals **(d)** bacteria and plants

6. A town draws more water from wells than rain recharges. In the long term: 1

(a) the water table falls **(b)** ozone increases

(c) the water table rises **(d)** wells overflow

7. Assertion (A): A convex lens can start a fire using sunlight. Reason (R): It converges parallel rays to a point. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

8. Assertion (A): The Moon is visible at night. Reason (R): The Moon produces its own light. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

9. Assertion (A): Decomposers are essential for the recycling of nutrients. Reason (R): They convert dead organic matter into simple inorganic substances. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. Assertion (A): The Moon has large changes in surface temperature between day and night. Reason (R): The Moon has a thick atmosphere. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

11. A clock's image in a plane mirror shows 10:10 in a mirror. The actual time is: 1

(a) 1:50 **(b)** 2:50 **(c)** 11:50 **(d)** 10:10

12. The new moon of 1 January 2024 is followed by a full moon after about: 1

(a) 15 days **(b)** 7 days **(c)** 60 days **(d)** 30 days

13. **Assertion (A):** Decomposers are essential in an ecosystem. 1
Reason (R): Decomposers break down dead plants and animals and return nutrients to the soil.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Weather events such as rain occur mostly in the troposphere. 1
Reason (R): The troposphere contains most of the air and water vapour of the atmosphere.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** A convex lens can be used to start a fire with sunlight. 1
Reason (R): A convex lens spreads out parallel rays of light.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do herbivores form the largest biomass in animals but fewer in number than plants? 2

17. Why are mountain areas colder than the plains even though they are nearer the Sun? 2

18. A person walks away from a plane mirror at 3 m/s. Find the speed at which the image moves relative to the person. 2

19. If a 1.5 m stick has a 2 m shadow, find the height of a building with a shadow 40 m long at the same time. 2

20. Why can heavy use of pesticides in fields harm the birds that feed on insects there? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A farmer sprays an insecticide that kills all insects in a field. Predict three consequences. 3

22. Estimate the percentage of the Earth's radius that is made up of the crust given that the thickest continental crust is 70 km and Earth's radius is 6400 km. 3

23. Explain the use of a reflector behind a bulb in a torch with a diagram description. 3

24. India is 82.5 degrees E for IST. Find the time difference between a place at 75 degrees E and IST. 3

OR

State how the afforestation of degraded land restores the ecosystem. 3

25. Explain how the biosphere depends upon the other three spheres with examples. 3

OR

If a mirror is curved outwards, how do you predict the nature of the image and its size? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** In a field in Punjab, wheat plants are eaten by grasshoppers, grasshoppers by frogs, frogs by snakes and snakes by eagles. Farmers killed many snakes, and soon the number of frogs increased and the number of grasshoppers fell sharply. 5

(i) Name the producer in this food chain.

(ii) Name the top carnivore.

(iii) Name the primary consumer.

(iv) Explain why killing snakes disturbed the balance of this ecosystem.

27. Analyse the food web: grass, grasshopper, rabbit, mouse, frog, snake, hawk. Identify all food chains, the trophic levels of the hawk, and the effect of removing the frogs. 5

OR

Analyse the data: Venus (460 degrees Celsius, thick CO2), Earth (15 degrees Celsius, thin atmosphere), Mars (-60 degrees Celsius, very thin). Explain the role of atmosphere and distance in these temperatures, and what Earth teaches about the greenhouse effect. 5

END OF QUESTION PAPER — **Total: 50 Marks**